

Motivation

Large language models can understand a household goal yet still fail to produce a sequence that is **syntactically valid, executable, and goal-completing**.

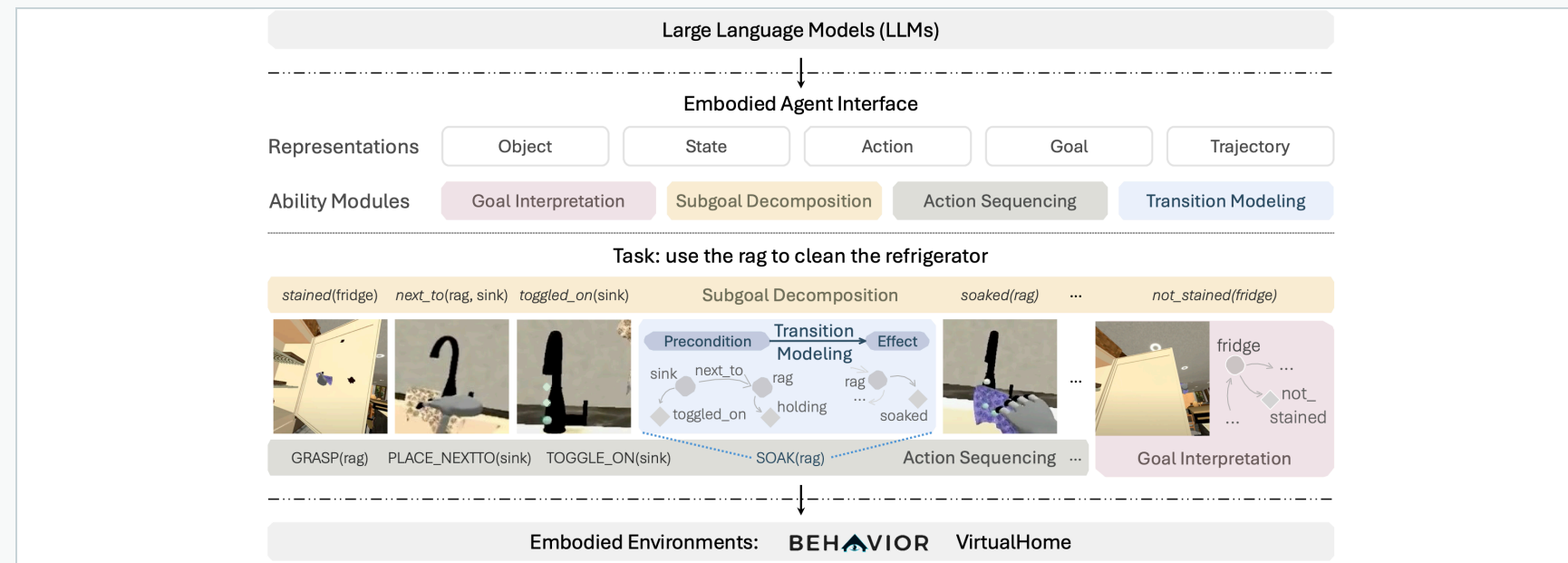
Many approaches bundle richer prompts, retrieval, validation and repair. When performance improves, it is unclear **where the gain came from**.

Under matched tasks and models, how much comes from better initial planning, how much from recovery, and how do the two interact?

Research Questions

- RQ1 — Grounding:** Do structured constraints, Flat RAG or GraphRAG improve initial plans?
- RQ2 — Failure:** Which trajectory and goal failures remain?
- RQ3 — Recovery & interaction:** How much does each recovery method add within each planner?
- RQ4 — Transfer:** Are effects consistent across three model families?

Benchmark & Scope



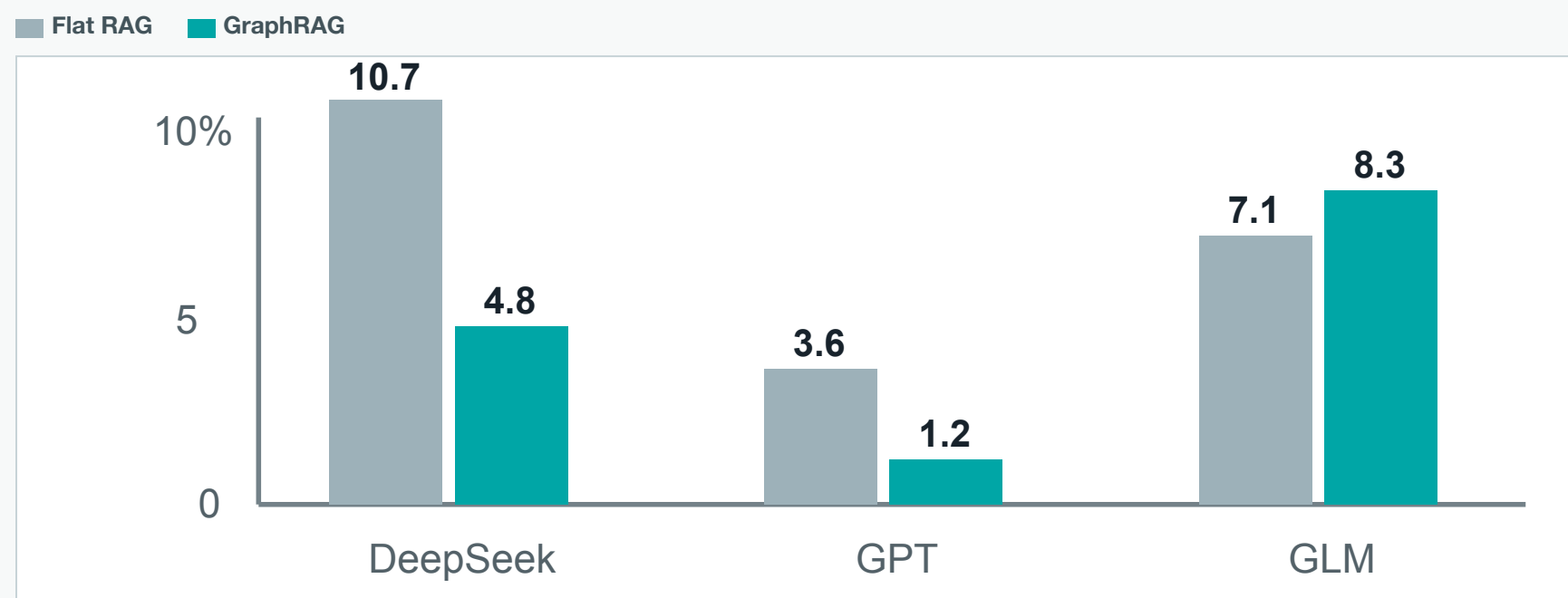
Embodied Agent Interface (EAI) links LLMs to symbolic representations and embodied environments. This study isolates VirtualHome Action Sequencing.

84 official-compatible tasks	8 task families
60 model-specific cells (two layers)	5,040 official records

Two evidence layers: a primary 4×4×3 B0–P1 factorial experiment (4,032 records) plus a matched 2×4×3 P1–P2 GraphRAG follow-up (2,016 records) on the same 84-task cohort. Shared P1 rows come from each layer's own run and are not interchanged. All predictions remain in the denominator.

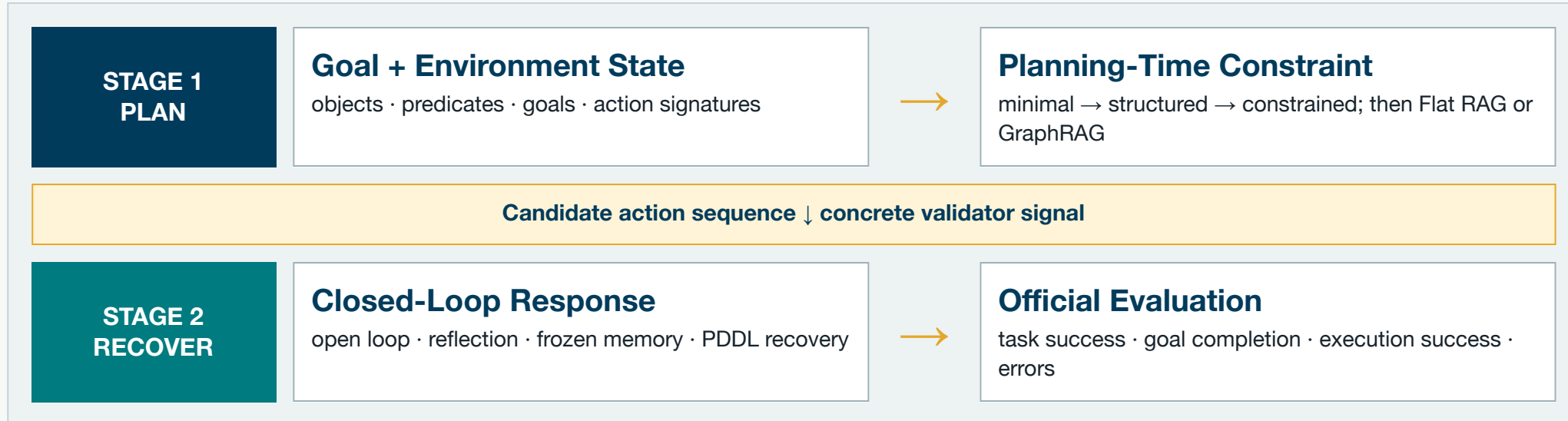
Why Model Effects Differ

Residual open-loop failures: Flat RAG → GraphRAG (% of 84 tasks)



GraphRAG reduced missing-step failures for every model, but introduced enough parsing failures on GLM to reverse its nettask-success gain.

Two-Stage Control Framework

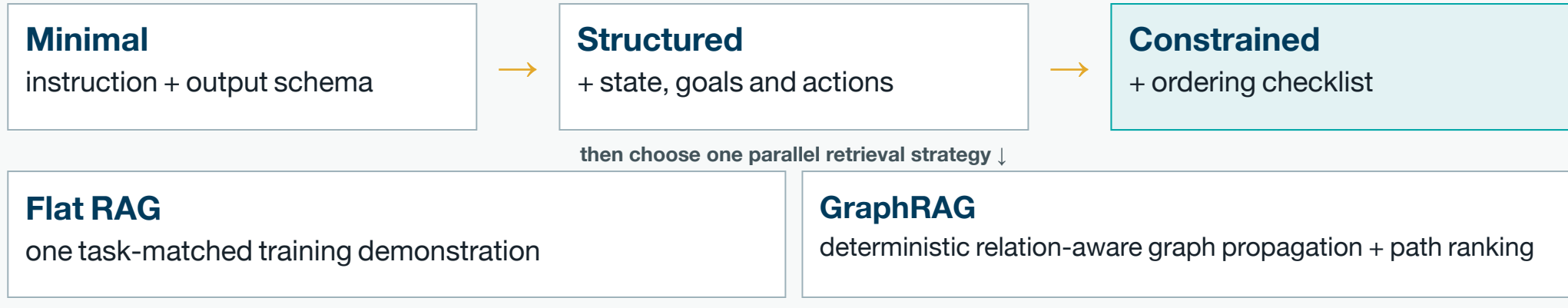


Separation principle: the local verifier is an intervention component. Only the pinned official VirtualHome evaluator produces reported outcomes.

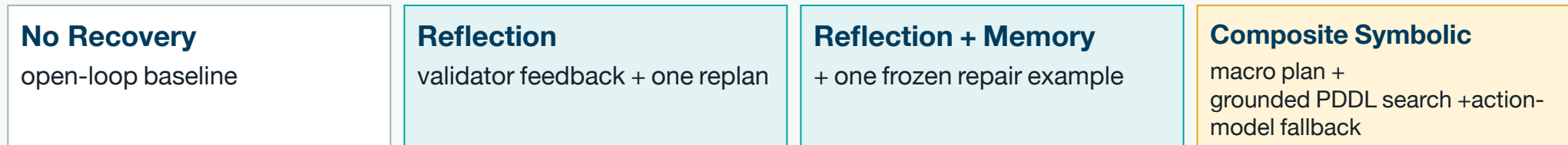
Treatments & Relationships

Two independent axes: choose one planning method and one recovery method for each experimental cell. Arrows mean “adds to”; branches mean alternative methods.

Axis A · Information available before the first plan



Axis B · Response after a detected failure



Reporting note: B0–P1 rows use the primary 4×4×3 factorial; the P2 GraphRAG row uses the matched 2×4×3 same-cohort follow-up.

Experimental Protocol

Matched design Primary 4×4×3 B0–P1 factorial plus a matched 2×4×3 P1–P2 GraphRAG follow-up on the same 84 tasks.	Models DeepSeek-V4-Flash, GPT-5.5 and GLM-5-Turbo.
Primary outcome Official task success rate; failures remain in the denominator.	Inference Family-clustered 95% CIs and exact paired McNemar tests.

Frozen evaluator: EAI commit 531c62f8. Temperature 0; each model uses the same 2,048-token output cap.

Threats to Validity

- Compatible subset, not the complete hidden EAI challenge set.
- Unseen task IDs but seen task families.
- VirtualHome Action Sequencing only; no BEHAVIOR result.
- GraphRAG is evaluated on this fixed cohort; generalisation requires a new compatible cohort.

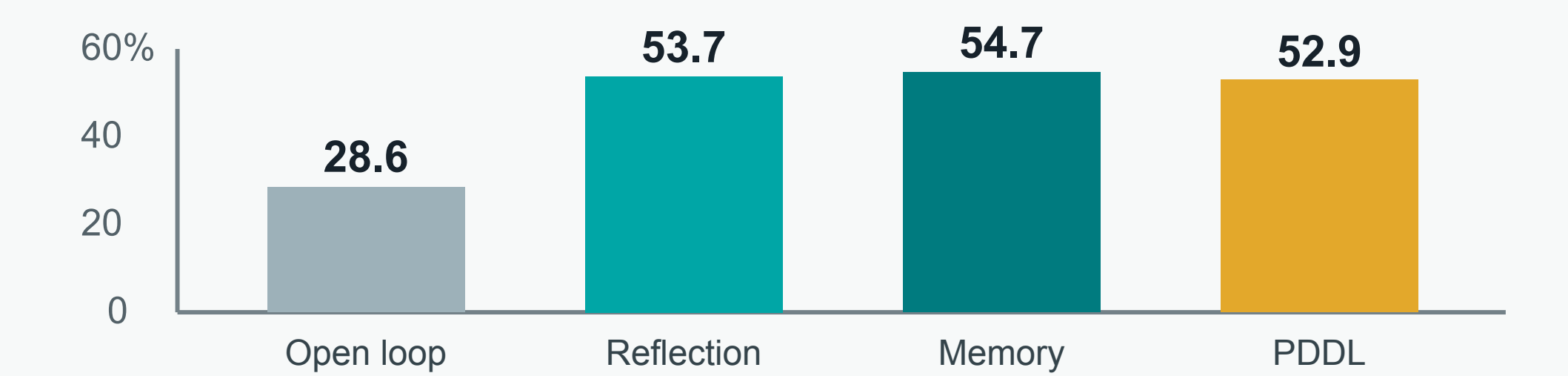
Selected References

- Xiang et al. *Embodied Agent Interface: Benchmarking LLMs for Embodied Decision Making*. arXiv:2410.07166, 2024.
- Puig et al. *VirtualHome: Simulating Household Activities via Programs*. CVPR, 2018.
- Shinn et al. *Reflexion: Language Agents with Verbal Reinforcement Learning*. NeurIPS, 2023.
- Lewis et al. *Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks*. NeurIPS, 2020.

Experimental Results

76.2% best task success GPT-5.5 · Minimal · Reflection	54.7% highest recovery mean frozen repair memory	99.8% PDDL mean execution success but only 52.9% task success
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Recovery dominates task success (mean across planners and models)



Task success is not the same as syntactic executability: symbolic recovery nearly eliminates execution errors but does not guarantee all task goals.

Planner × Recovery Interaction

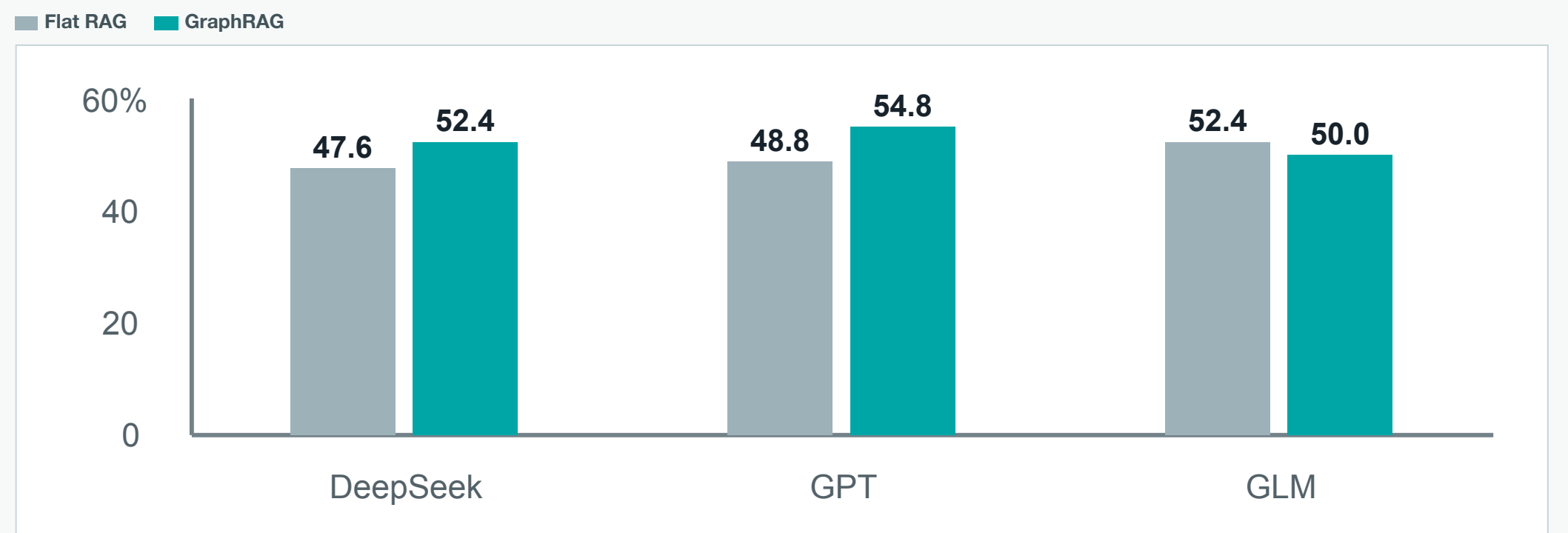
Official task success (%) averaged across three models — B0–P1 primary layer; GraphRAG row from matched P1–P2 follow-up

	Open loop	Reflect	Memory	PDDL
Minimal	0.0	61.1	64.7	52.4
Structured	22.6	53.2	52.4	53.6
Constrained	18.3	50.8	52.0	54.8
Flat RAG	49.6	50.8	50.8	50.8
GraphRAG	52.4	52.8	53.6	53.2

Diminishing returns: recovery adds up to +**76.19 percentage points** to an ungrounded baseline, but only 0–3.57 points once retrieval-based planners already produce strong executable plans.

Model-Specific Planning Effect

Open-loop task success: Flat RAG vs GraphRAG (%) — matched P1–P2 follow-up



GraphRAG gains were model-dependent: +4.76 pp DeepSeek, +5.95 pp GPT, but –2.38 pp GLM.

Conclusion

Where intervention occurs matters as much as which component is used.

Planning-time retrieval improves first-pass plans; execution-time recovery is most valuable when those plans are weak. The study shows that planning and recovery must be evaluated together rather than as isolated components.